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Economic evaluation of rooftop grid-connected photovoltaic systems for residential building in Egypt

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Summary

Egypt is a country with high solar energy potential and the exploitation of such promising energy resource is critical for national sustainable development through efficient energy planning with gradual independence from fossil fuels. Successive incentive polices had been introduced by the Egyptian electricity authority to encourage the deployment of small-scale residential rooftop photovoltaic (PV) systems. This article studies the techno-economic feasibility of grid-connected rooftop PV system in Egypt under the currently applied retail electricity price and the net energy metering policy. The study investigates three types of residential households with different electricity demand levels; low, medium, and high consumptions. The economic evaluation of various sizes of the PV system is carried out based on different economic measures such as net present value, cost of energy, payback period, and electricity bill saving. Hybrid Optimization of Multiple Energy Resources software has been utilized to carry out such economic evaluation. The results identify that the viability of PV installation in residential applications is clearly affected by the energy consumption pattern, the parameters of the incentive policy applied, and the economical indices of the system.

KEYWORDS

grid connected PV, net energy metering, residential load, rooftop PV system

1 | INTRODUCTION

Energy scarcity and climate change are urgent issues worldwide led to the development of renewable energy resources to achieve a sustainable energy system. Among all renewable energy resources, solar photovoltaic (PV) have received much attention due to their economic and environmental benefits.[1] The PV sector has demonstrated significant progress in recent years, reaching more than 505 GW of installed capacity in 2018. China (176.1 GW), United States (62.4 GW), Japan (56 GW), Germany (45.3 GW), Italy (20 GW), and India (32.9 GW) cover about 77% of the global solar PV power installed.[2]

List of Symbols and Abbreviations: AC, alternative current; $C_{ann,tot}$, annualized total cost; CO_2, carbon dioxide; COE, cost of Energy; C_0, total initial investment cost; C_t, net cash inflow during the period t; CPP, critical peak pricing; $C_{NPC,tot}$, total net present cost; CRF, capital recovery factor; DC, direct current; E_{served}, total electric load served; FiT, feed in tariff; IRR, internal rate of return; NEM, net energy metering; NPV, net present value; PB, payback period; PV, photovoltaic; R, project lifetime; r, discount rate; RF, renewable fraction; RTP, real time pricing; T, project total period; t, number of time periods; TOU, time of use.

Integrating PV systems with the utility grid is considered a way to reduce the energy cost and consumption.[3] In fact, the price of residential PV system is decreased of about 80% from 2008 to 2012 in most competitive markets.[4] There are a lot of factors and conditions that may affect the PV adaptation in different countries like, weather and climate conditions, industrial capabilities, incentive policies, and the tariff structure for the grid-connected PV generation.

This article studied the economic evaluation of rooftop grid-connected PV system for residential buildings in Egypt. The economic parameters used as input to the studied model are the actual present parameters obtained from latest official Egyptian economic references.

Many researches had been implemented to evaluate the viability of PV integrating for the residential purposes to the utility grid by different techniques.[5-26] Different researches used more than one parameter to evaluate the integration of PV system with the utility grid by using different software techniques. The net present value (NPV), internal rate of return (IRR), and the payback period (PB) were the most common parameters which had been used for different systems evaluation.

Unlike previous researches concerning the same point of study this article considers the effect of discount rate variation on the evaluation study results and show the subsidy releasing influence on such results. The article also takes into account different energy consumption patterns to clarify its effect on the profitability of PV system.

2 | LITERATURE REVIEW

Recently, in 2019, a case study in Morocco by A. Allouhi, et al, studied the effect of three different PV cell technologies (Polycrystalline, Monocrystalline, and amorphous on Microcrystalline) on the cost of Energy (COE) and the simple payback period using PVSyst software with different PV system capacities under constant electricity rate. It showed that, the minimum COE and the shortest payback period was obtained by the polycrystalline cell technology, while the maximum COE and the longest payback period was obtained by the amorphous on Microcrystalline cell technology. But it did not take into consideration neither the effect of the load profile nor the pricing policy between the customers and the government.[5] Duong, et al[6] implemented an optimization technique to reduce the power loss and harmonics of a PV system using the optimum PV size and location to comply with IEEE Std. 519. Angel Vargas, et al., used the previous parameters to evaluate a system with different PV sizes according to the Paris agreement. It showed that, according to the current price of the PV and the electricity rate, the using of the residential PV is profitable for all analyzed locations, while this article did not include the effect of the sellback price of the excess PV power to the grid on the profitability on the system.[7] Also, the same parameters were used for another evaluation in Palestine by using PVSyst software under feed in tariff (FiT) and net energy metering (NEM) dynamic pricing policy. It was found that, the payback period of the system is about 5 years. Although, Palestine do not have a PV industry and it imports all the system equipment from a foreign country, using the PV in the residential sector is economically feasible due to the low price of the PV modules.[8]

The same software was used but with implementing variable electricity tariffs and the system evaluation was from the COE point of view.[9] This study did not consider the program implementation costs and it focused only on the incentive's analyses. The dynamic pricing schemes had been illustrated in an Indian study in order to reach the minimum electricity bill under the NEM pricing policy.[10] It studied the effect of different tariff types on the bill savings under the NEM policy. It was concluded that, the bill savings decreases by the increasing of the PV penetration for all dynamic pricing schemes. According to the increase of using different software in the evaluation process, some of these had been evaluated. It was found that, PV SOL software to be the easiest, fastest, and the most reliable software for the PV system simulation. While it does not consider neither the pricing policy nor the tariff scheme for system evaluation.[11] The NPV and the IRR was used to study the system viability as an evaluation parameter.

In Thailand,[12] used a constant PV system size with constant electricity tariff to achieve the appropriate FiT policy for the residential customer, and it was found that investing in the residential rooftop PV system is not economically feasible under the present FiT scheme and decreasing the current market price by 35% may reach the expected investment return. While, different researchers used a variable system sizes with variable electricity tariffs to get the optimum PV system design with the best policy for different countries.[13,14] Both used the IRR as a parameter for evaluation process. Also, a variable system capacity with different load patterns was used to evaluate the different Italian electricity pricing policies.[15] It was noticed that for the current tax program obtains the shorter payback period. But actually, it is not feasible that the payback period depends on the site latitude.

In Greece, the Benefit-Cost Ratio (BCR) with the assistance of RET Screen Software was used as an indicator to prove that the Greece market needs more incentive policies from the Greek government to promote PV installations.[16] The energy yield and the energy performance ratio were used as an evaluation parameter to get the best PV technology, but it did not include neither the dynamic pricing policy nor the different tariff schemes.[17] Two Norwegian Studies had been implemented in 2015, the first was aimed to achieve the best system design under variable system size and the NPV as an evaluation parameter,[18] it was concluded that, according to the low financial support from the government, the rooftop PV system is not feasible. While the second one used the COE to find the barriers and difficulties for investing in PV systems and it was suggested; (a) Provide an attractive FiT system to encourage investing in the residential PV system. (b) Provide about 40% of the initial investment cost as an incentive investment.[19] Another way of evaluation was implemented by introducing a review for the different policies and situations in different countries and ends with a lot of suggestions to promote and encourage using of PV systems.[20-22] Ayompe et al.[23] used a constant system size, constant electricity tariff, and constant load profile under the FiT pricing policy to evaluate the system cost by means of the NPV in Ireland and it was found that the grid parity will occur on 2016 and a positive NPV will be achieved on 2021.

In Reference 24, Homer software was used to evaluate the effect of adding rooftop PV system to the utility grid, and it was found that using PV systems may reduce the electricity cuts especially in summer, on the other hand it was noticed that the PV cells efficiency reduces during the summer period according to the temperature increase. Reference to,[25] it was concluded that, consuming a small amount of power from the grid will lead to a minimum electricity tariff, while it is not guaranteed to get the minimum tariff by implementing minimum power from grid, because a small quantity of power may be consumed in a high peak period with high rate. In Reference 26, it was showed that, only 0.1% of the Brazilian residence would be ready for using PV system in 2016. This ratio will increase to be 55% in 2026, while this study assumed that all economic factors is logical which needs more studies to evaluate the different opportunities for the Brazilian customers. From the previous review it is shown that there is a lack in the research from the pricing scheme point of view beside the pricing policy between the customer and the utility grid. The overall evaluation of the aforementioned research studies is represented in Table 1.

The rest of article is organized as follows, the Egyptian current situation case study in Section 2. The economic evaluation parameters are presented in Section 3, followed by Site Selection, Electrical Load, and Energy System Components is presented in Section 4. Section 5 discusses the work and gives pointers for the evaluation process followed by parameters sensitivity analysis in Section 6. Finally, the article is concluded in Section 7.

3 | EGYPTIAN CASE STUDY

In Egypt, according to the latest annual report which had been released from Egyptian Electricity Holding Company (EEHC) on 2018, the total installed generation capacity is about 54.5 GW. Only 10% of that total is gained from renewable energy including the large scale hydrogeneration.[27]

Dealing with the economic development, Egypt have set ambitious target to achieve 42% of its electricity generation capacity from renewable energy sources by 2035 as shown in Figure 1.[28]

The solar atlas of Egypt is shown in Figure 2.[29] From the solar atlas it is clear that Egypt has a high solar energy and high solar radiation ranges from 1950 kWh/m^2/yr on the Mediterranean coast to more than 2600 kWh/m^2/yr in upper Egypt, while about 90% of the Egyptian territory has an average global radiation greater than 2200 kWh/m^2/yr.[29]

The utilization of using these potentials is critical for the sustainable development in Egypt and to reduce the dependency on fossil fuels. The total installed PV capacity in Egypt is about 300 MW.[28] Only 30 MW of which are grid-connected rooftop PV systems. From 2013, Egypt started to introduce incentive policies to encourage implementations of small- scale PV systems, especially with annual increase in retail electricity price. The effectively running net metering policy started in September 2017 reached a sell price of $0.04/kWh. Summary of the implemented incentive policies are shown in Table 2. The Egyptian current electricity tariff is based on a block rate structure as shown in Table 3.

4 | DATA AND METHODOLOGY

The case study site used in this research is a typical residential multi villas compound near to Alexandria governorate in northern Egypt, located on 31° 4.0′ N and 29° 43.8′ E. The scaled annual average solar radiation at such location is 5.2 kWh/m^2/day.

TABLE 1 Summary of grid-connected rooftop residential PV system evaluation techniques

No.	Evaluation parameters	Variable parameters	PV capacity	Purchasing cost ($/kWh)	Load profile	Tariff	Incentive policy	Software	Country	Date	References
1	PV technology, PB, COE	PV cell technology, system size	Var.	Const.	—	Fixed	—	PV SYST	Morocco	2019	[5]
2	NPV, IRR, PB	System size	Var.	Const.	Const.	Fixed	—	MatLab	France, Spain	2018	[7]
3	NPV, IRR, PB	Policy, electricity rate	Const.	Var.	—	Fixed	FiT, NEM	PV SYST	Palestine	2018	[8]
4	COE	Policy, electricity rate	Const.	Var.	Const.	Fixed	FiT, NEM	PV SYST	Brunei	2018	[9]
5	Bill saving	Tariff scheme, System size, Electricity rate	Var.	Var.	Const.	Flat Rate, CPP, RTP, TOU	NEM	MatLab	India	2018	[10]
6	Software evaluation	Software	Const.	Const.	Const.	Fixed	—	PV SOL PVGIS SolarGIS ISIFO	India	2017	[11]
7	NPV, IRR	Policy	Const.	Const.	—	Fixed	FiT	RET screen	Thailand	2017	[12]
8	IRR	Policy, System size, Electricity rate, Load profile	Var.	Var.	Var.	Fixed	FiT	MatLab	Austria	2017	[13]
9	NPV, IRR, PB	Policy, System size, Electricity rate	Var.	Var.	—	Fixed	FiT	RET Screen	Different countries	2016	[14]
10	NPV, PB	System size, Load profile	Var.	—	Var.	Fixed	—	MatLab	Italy	2017	[15]
11	NPV, BCR, IRR	System size	Var.	—	—	Fixed	—	RET screen, SimaPro	Greece	2017	[16]
12	Energy yield and performance ratio	Tilt angle	Const.	—	Const.	Fixed	—	SolarGIS	India	2016	[17]
13	NPV	System size	Var.	—	—	Fixed	—	PV SYST	Norway	2015	[18]
14	COE	Policy	Const.	Const.	Const.	Fixed	FiT	MatLab	Norway	2015	[19]
15	NPV, IRR, PB	System size, Electricity rate	Var.	Var.	—	Fixed	FiT	MatLab	Thailand	2015	[20]
16	NPV	Policy	Const.	Const.	Const.	Fixed	FiT	MatLab	Ireland	2010	[23]
17	NPV	None	Const.	—	Const.	Fixed	—	Homer	Algeria	2014	[24]
18	COE	System Size	Var.	—	—	Fixed	—	SAM	Brazil	2015	[26]
19	NPV, COE, Bill savings, Annual energy purchased, Annual energy sold, and Renewable fraction	PV size, Load profile	Var.	Var.	Var.	Var.	NEM	HOMER	Egypt	2019	Present work

Abbreviations: COE, cost of Energy; CPP, critical peak pricing; FiT, feed in tariff; IRR, internal rate of return; NEM, net energy metering; NPV, net present value; PB, payback period; PV, photovoltaic; RTP, real time pricing; TOU, time of use.

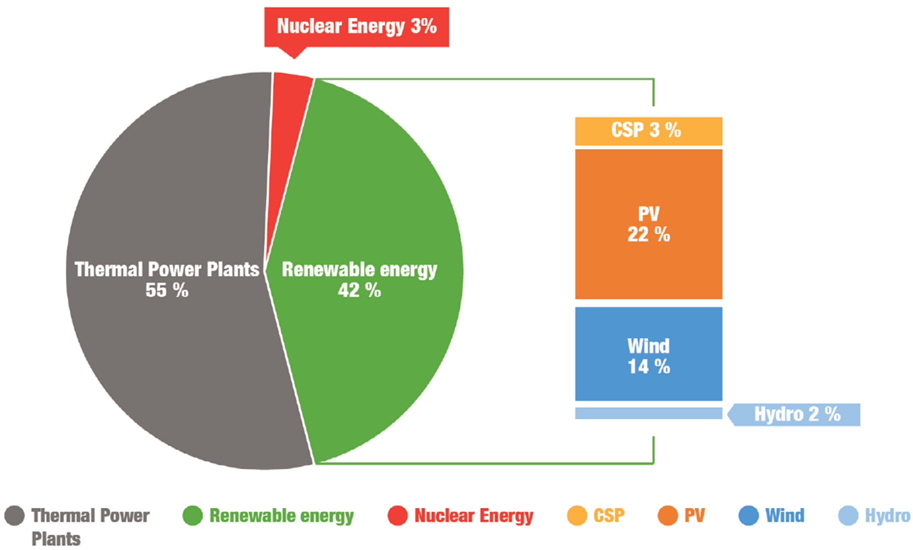

FIGURE 1 Egyptian electricity target generation in 2035[28]

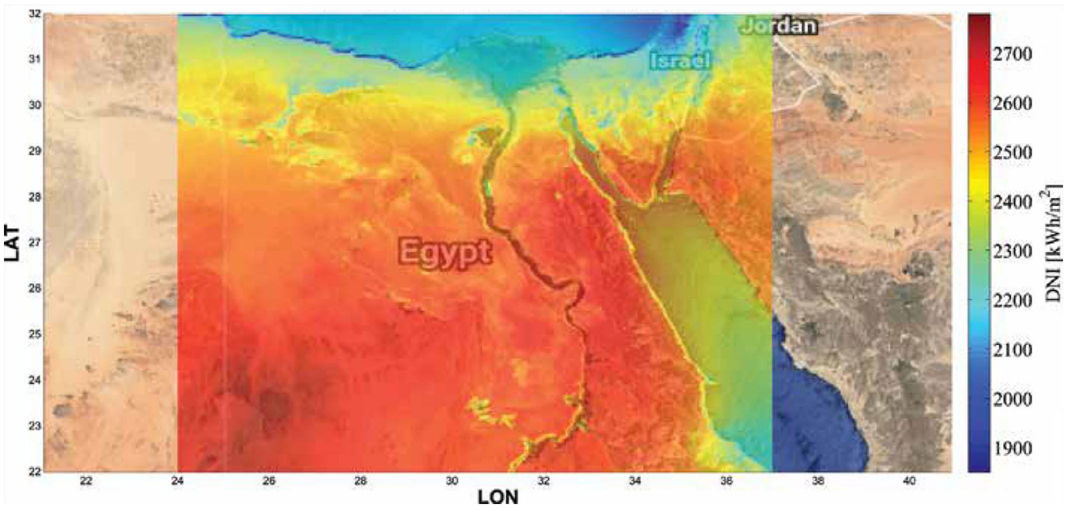

FIGURE 2 Egypt global horizontal solar irradiation atlas[29]

TABLE 2 Announced incentive policies in Egypt[30]

Date	Announcement	Tariff
2013	Begin using the NEM system	Equal to the highest block rate
Oct 2014	Begin using the FiT system	\$0.047/kWh
Sep 2016	Update of using the FiT system	\$0.058/kWh
Mar 2017	Begin using the NEM system	\$0.039/kWh
Sep 2017	Update of using the NEM system	\$0.04/kWh

Abbreviations: FiT, feed in tariff; NEM, net energy metering.

TABLE 3 Egyptian block rate tariff structure[30]

Level (kWh)	Consumption (kWh)	Cost (\$/kWh)
0-100	0-50	0.012
	50-100	0.017
101-1000	0-200	0.020
	201-350	0.039
	351-650	0.05
	651-1000	0.076
Above 1000	0 to more than 1000	0.081

The Egyptian daily load profiles for three different load patterns; low, medium, and high consumption are shown in Figure 3. The daily energy consumption for these patterns are 18.8, 28.9, and 35.28 kWh/day respectively. For the three different load patterns, the load was assumed to be increased by 20% during the peak times, (from 5:00 PM to 9:00 PM) in summer season.[31]

The grid-connected rooftop PV system, as shown in Figure 4, consists of PV panels, inverter and the utility grid. The current incentive policy which is used to encourage residential loads to use grid-connected PV system is NEM. The PV output power primary supplies the demand load, while the excess remaining power from the PV panels is sold back to the utility grid. If the load demand exceeds the PV output the rest of the load power has to be supplied by the utility grid.

Based on a local market survey, five different sizes of PV units are considered in this article which are 5, 7, 10, 15, and 20 kW. Each PV unit has its capital, replacement, operation, and maintenance costs as shown in Table 4, and has 25 years lifetime. The Inverter sizing was varying according to the PV system size and each inverter is 10 years lifetime and has its capital, replacement, operation, and maintenance costs as shown in Table 5. The overall project lifetime is 25 years.

The economic feasibility of grid-connected PV power system for residential load demand is carried out using the following economical parameters; NPV, COE, IRR, and payback period (PBP). The detailed definitions of such economical parameters are presented as per reference[32];

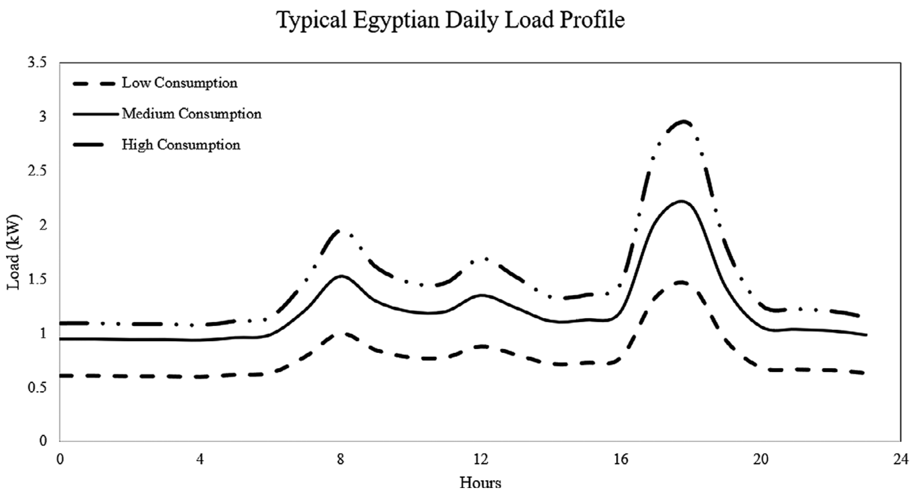

FIGURE 3 Egyptian daily load profiles for three different load patterns

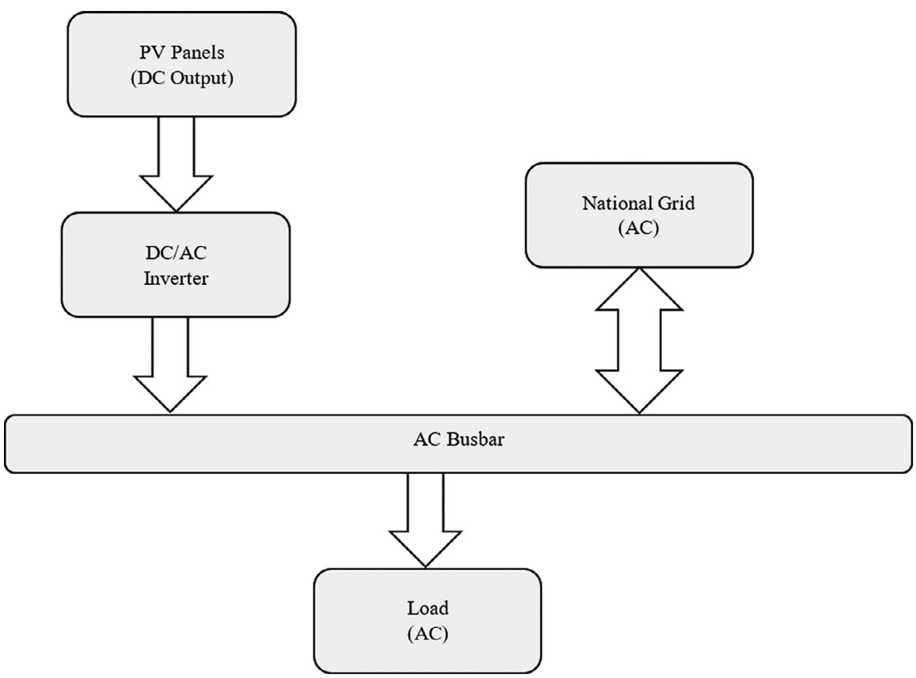

FIGURE 4 Basic topology of the grid-connected PV system

TABLE 4 PV System different sizes and its related costs

PV system size (kW_p)	Capital cost ($)	Replacement cost ($)	Operation and maintenance. cost ($)
5	2070	1450	20
7	2830	1980	
10	3920	2750	
15	5660	3960	
20	7300	5110	

Abbreviation: PV, photovoltaic.

TABLE 5 Inverter different sizes and its related costs

Inverter size (kW_p)	Capital cost ($)	Replacement cost ($)	Operation and maintenance cost ($)
5	1005	705	30
7	1330	930	
10	1686	1180	
15	1900	1330	
20	2233	1563	

NPV is the difference between the present value of cash inflows and the present value of cash outflows. It is used in to analyse the profitability of a projected investment or project.

$$\mathrm{NPV} = \sum_{t=1}^{T} \frac{C_t}{(1+r)^t} - C_0 \tag{1}$$

where C_t is the net cash inflow during the project lifetime t, and it means the difference between the project inflow and outflow to represent the amount of money produced or lost during the project period. Without having a long-term positive cash net flow, the project will fail and will be unprofitable. C_o is the total initial investment costs, or the amount required to start the project. It is equal to the capital expenditures plus working capital requirement. Smaller investment cost will lead to a higher NPV and more project profits. r is the discount rate. t is the number of time periods. *COE* is the average cost per kWh of useful electrical energy produced by the system, it is a meter to compare systems.

$$\mathrm{COE} = \frac{C_{ann,tot}}{E_{served}} \tag{2}$$

where $C_{\mathrm{ann,tot}}$ is the annualized total cost and E_{served} is the total electric load served.

$$C_{ann,tot} = CRF(r,R).C_{NPC,tot} \tag{3}$$

$$\mathrm{CRF}(r,R) = \frac{r(1+r)^R}{(1+r)^R - 1} \tag{4}$$

where CRF is the capital recovery factor and R is the project lifetime.

$C_{\mathrm{NPC,tot}}$ is the total net present cost, it represents the value of all costs of the system (includes Capital, Replacement, and Maintenance Cost) minus the value of all revenue earned over its lifetime.

IRR is used to measure the profitability of potential investments and it is also the discount rate at which the NPV equal to zero.

PBP defined as the length of time required to recover the cost of an investment. The payback period ignores the time value of money unlike the IRR.

In this article, Hybrid Optimization of Multiple Energy Resources (HOMER) software was used as the calculation tool for all the previous economic parameters. HOMER uses a derivative free optimization algorithm that select the optimum system component values from a predetermined search space to get the minimum Cost of Energy (COE) and Net Present Cost (NPC).[33]

5 | RESULTS AND DISCUSSION

In the following simulation results the grid power purchasing price is different for the three load patterns, as explained in Table 3; \$0.05/kWh, \$0.075/kWh, and \$0.081/kWh for low, medium, and high energy consumption, respectively. The grid net excess price is constant with \$0.04/kWh for all energy consumptions scenarios. Depending on the reports and the statistics gained from the central bank of Egypt, the discount rate used in the study is 17.25%[34] and the inflation rate is 17.68%.[35]

5.1 | Low energy consumption pattern

The base case for this system is to feed the load (18.8 kWh/d) completely from the grid without providing the PV system, with \$0.05/kWh power price and \$0.04/kWh sellback price, and it was found that, the COE is \$0.05/kWh and the NPV is \$-8995.

Installing rooftop PV system will increase the NPV and decrease the COE.

The effect of different PV sizes on the economic indicators was varying, as shown in Figure 5.

Figure 5A showed that, the NPV is increasing with the size of the PV system. The installed PV system will be profitable when installing more than 7.8 kW_p PV, size as it is the turning point from the loose to the profit. At this point the NPV and the COE is equal to zero, after which the profit will increase with larger PV sizes. This result is complying with Reference 16 which indicated that, installing more than 5 kW_p will increase the system profitability. On the other hand, the energy cost is decreased rapidly by increasing the PV size because the system reduced the dependency on the utility grid, and it depends on the PV. The negative cost of energy means that a high-power generation meets a low demand.

Figure 5B showed that, increasing the PV size for this load pattern, will increase the amount of energy sold to the utility and will reduce the electricity bill hence increases the bill savings.

Installing more than 10 kW_p PV size for this load pattern will not have any effect on the annual energy purchased from the grid because the load demand is small compared to the PV size, while installing larger PV sizes will lead to a rapid increase in the annual energy sold to the grid that will lead to sell more power to grid and gain more financial profits because the PV supplies a small quantity of load and there is a considerable surplus power to be sold. Increasing the PV size will lead to more share of the renewable energy with the grid till it reaches about 91% for 20 kW_p PV size, and became near to achieve the self-sufficiency as shown in Figure 5C.

5.2 | Medium energy consumption pattern

The base case for this system is to feed the load (28.9 kWh/d) completely from the grid without providing the PV system, with \$0.075/kWh power price and \$0.04/kWh sellback price, and it was found that, the COE is \$0.075/kWh and the NPV is \$-20 743.

Installing rooftop PV system will increase the NPV and decrease the COE.

The effect of different PV sizes on the economic indicators was varying, as shown in Figure 6.

Figure 6A showed that, the project will be profitable when installing 10.7 kW_p PV size as it is the turning point from the loose to the profit, the NPV and the COE is equal to zero, and the profit will increase by increasing the PV size. On the other hand, the cost of energy decreased rapidly by increasing the PV size because the system reduced the dependency on the utility grid, and it depends on the PV.

Figure 6B showed that, increasing the PV size for this load pattern, will share in reducing the energy purchased from grid, that will increase the amount of energy sold to the utility and will reduce the electricity bill hence increases the bill savings.

Installing large sizes of PV will not have any effect on the annual energy purchased from grid. Installing 5 kW_p PV for this load pattern will save about 4 MW yearly while the difference between 5 and 20 kW_p is about 1 MW yearly.

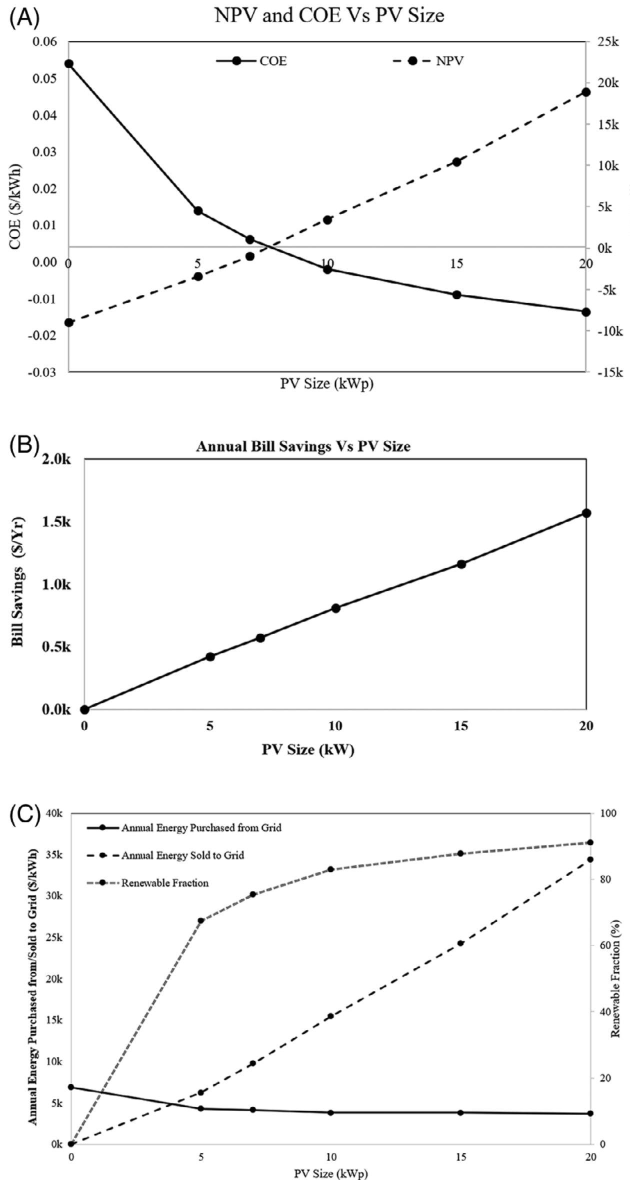

FIGURE 5 A, Effect of PV size on NPV and COE or low energy consumption pattern. B, Effect of PV size on the annual bill savings for low energy consumption pattern. C, Effect of PV size on the annual energy purchased, sold, and the renewable fraction for low energy consumption pattern

Increasing the PV size is perfect from the energy sold to grid point of view as it will feed the utility with the surplus power after feeding the load.

As mentioned before, increasing the PV size will lead to more share of the renewable energy with the grid, but for this load pattern the renewable sharing will be less than the low load pattern and it will reach about 86% after installing 20 kW_p PV system size as shown in Figure 6C.

5.3 | High energy consumption

The base case for this system is to feed the load (35.28 kWh/d) completely from the grid without providing the PV system, with $0.081/kWh power price and $0.04/kWh sellback price, and it was found that, the COE is $0.081/kWh and the NPV is $-27 734.

Installing rooftop PV system will increase the NPV and decrease the COE as well.

The effect of different PV sizes on the economic indicators was varying, as shown in Figure 7.

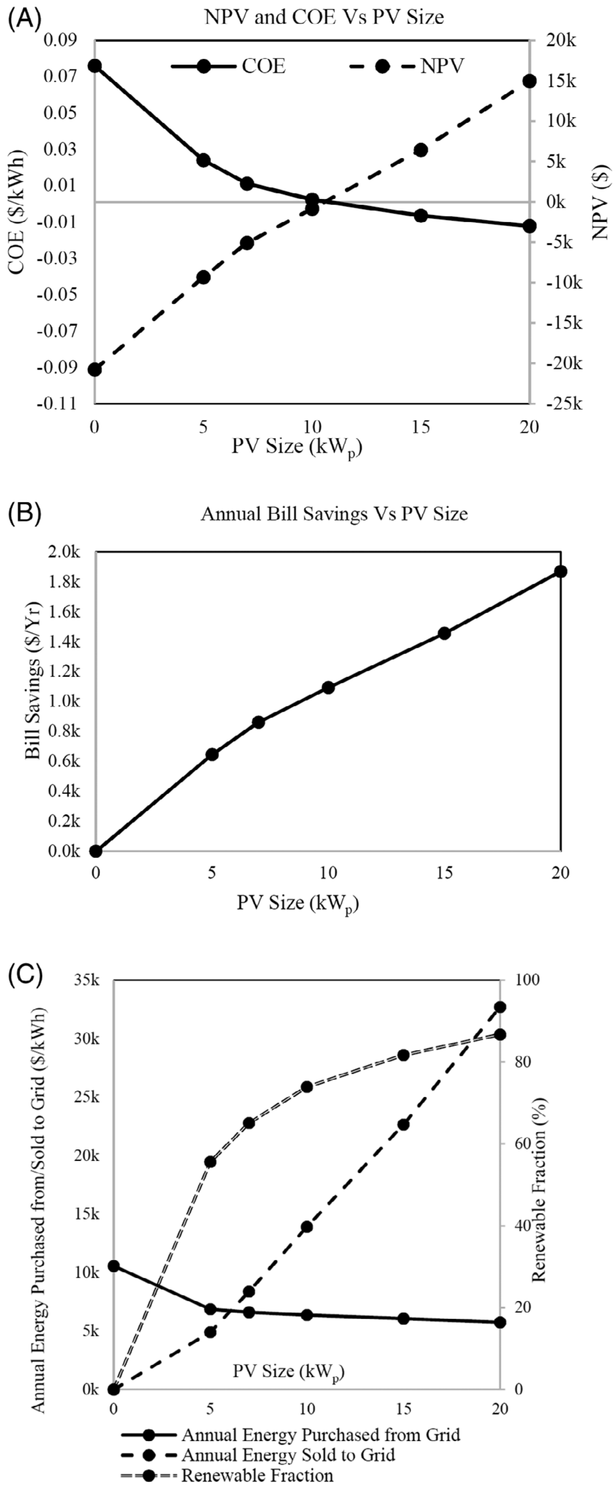

FIGURE 6 A, Effect of PV size on NPV and COE. B, Effect of PV size on the annual bill savings. C, Effect of PV size on the annual energy purchased, sold, and the renewable fraction

Figure 7A showed that, the project will be profitable when installing 12.87 kW_p PV size as it is the turning point from the loose to the profit, the NPV and the COE is equal to zero, and the profit will increase by increasing the PV size. On the other hand, the cost of energy decreased rapidly by increasing the PV size because the system reduced the dependency on the utility grid, and it depends on the PV.

Figure 7B showed that, increasing the PV size for high load pattern, will increase the amount of energy sold to the utility and will reduce the electricity bill and increase the savings due to the big difference between the purchasing price and the sellback price.

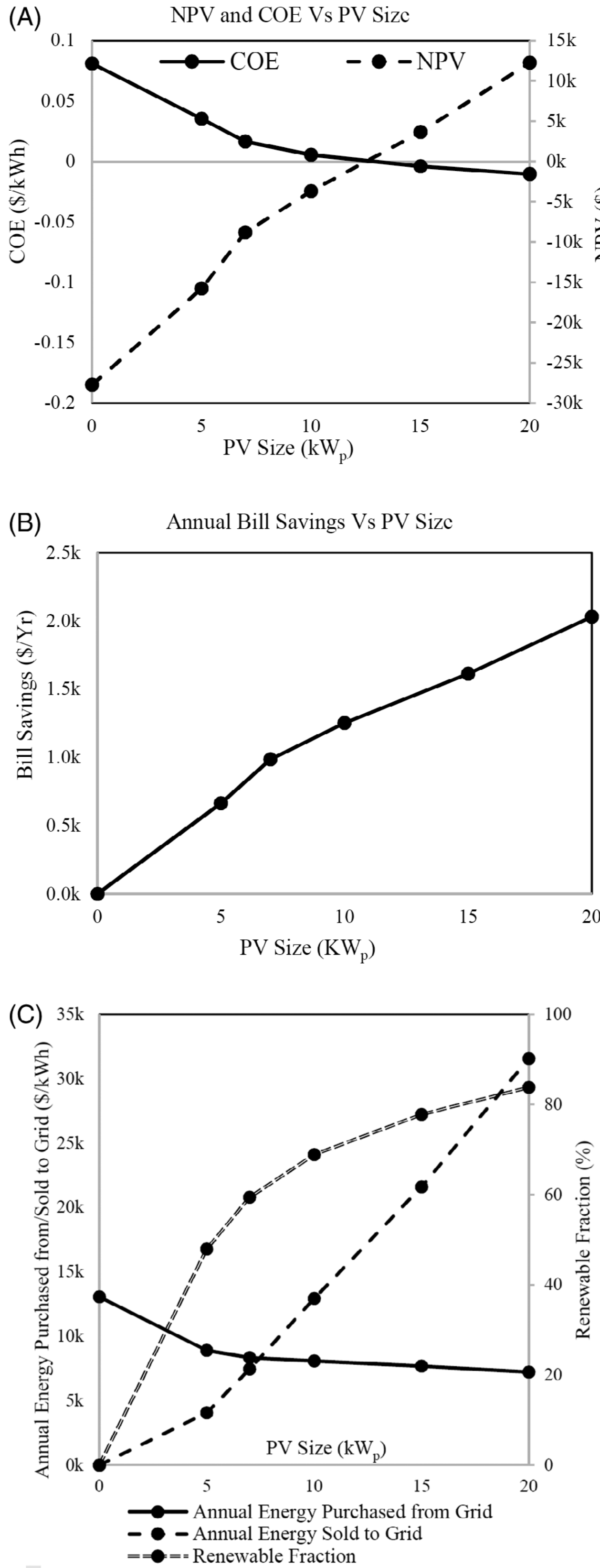

FIGURE 7 A, Effect of PV size on NPV and COE. B, Effect of PV size on the annual bill savings. C, Effect of PV size on the annual energy purchased, sold, and the renewable fraction

Installing more than 7 kW_p of PV will not have any effect on the amount of energy purchased from the grid annually as the difference between 7 and 20 kW_p is about 1.3 MW yearly.

Definitely, Increasing the PV sizes will increase the amount of energy sold to the grid after feeding the load but with smaller scale than the previous load patterns because of high load consumption.

According to the high load consumption, increasing the PV sizes will also increase the renewable share with the grid but with smaller scale than the previous load patterns as it reached about 83% after installing 20 kW_p PV system as shown in Figure 7C.

6 | SENSITIVITY ANALYSIS

The system assumptions at which all the above economical indices results are obtained may be subject to variations. The main varying parameters that will be considered for system sensitivity analysis are sellback power price, releasing electricity tariff subsidy, the discount rate and PV system capital cost.

First: the sellback power price was varying as follows ($0.04, 0.05, 0.07, 0.08, and 0.1/kWh), as shown in Figures 8, 12 and 16.

Second: Egyptian electricity tariff is subjected to the means of subsidy for different load patterns. The effect of releasing subsidy had been studied. The real electricity rate without subsidy for low energy consumption load pattern is $0.123/kWh instead of $0.05/kWh, for medium energy consumption load pattern is $0.141/kWh instead of $0.075/kWh, while the high consumption load pattern does not have any subsidy. The electricity tariff rate was varying for load pattern with different rates to gradually release the subsidy, as shown in Figures 9, 13 and 17. These variations aim to study its effect on the NPV for the proposed three load patterns.

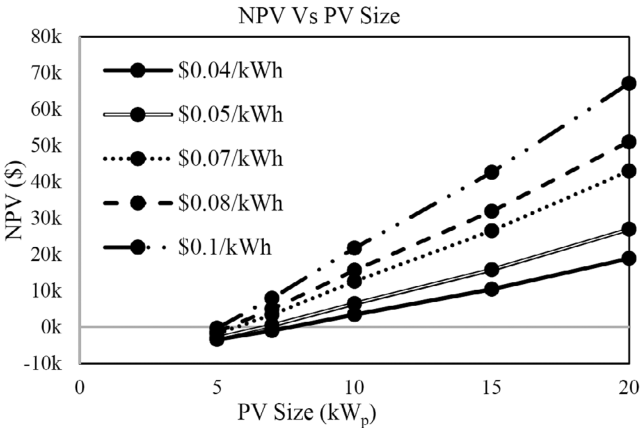

FIGURE 8 Effect of different sellback prices on the NPV

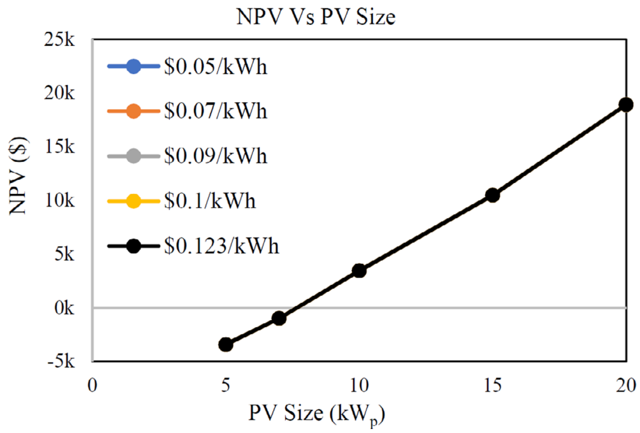

FIGURE 9 Effect of different sellback prices on the NPV

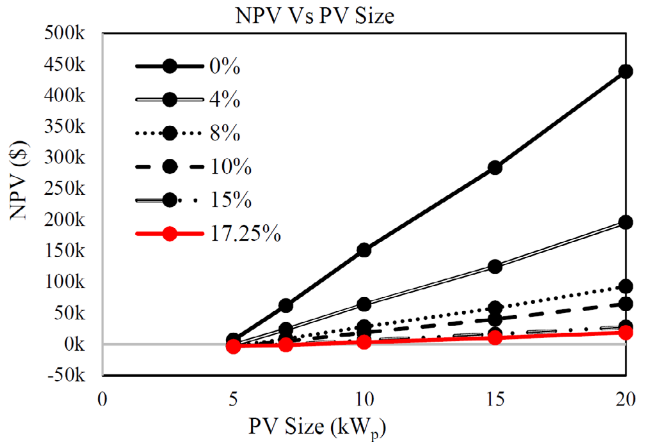

FIGURE 10 Effect of different sellback prices on the NPV

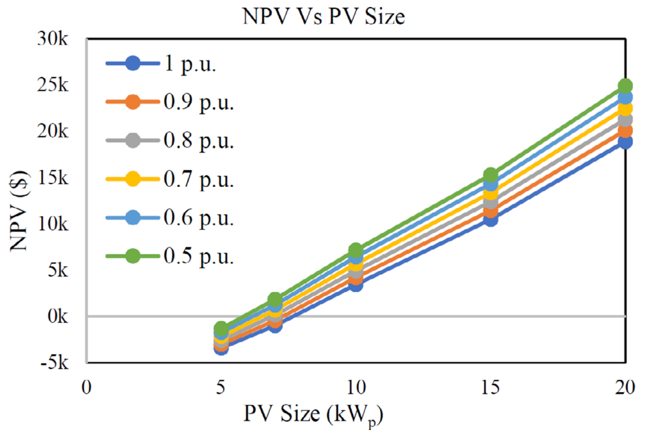

FIGURE 11 Effect of power purchasing prices on the NPV

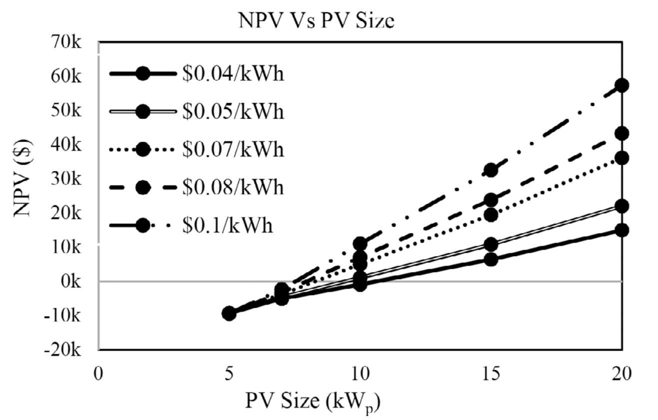

FIGURE 12 Effect of power purchasing prices on the NPV

Third: the discount rate was varying as follows (0, 4, 8, 10, 15, and 17.25%), as shown in Figures 10, 14 and 18.

Fourth: The PV system capital cost has fallen rapidly over the past 10 years and is expected to continue declining in the future that makes the households interested in installing and investing in the PV. This reduction in the PV system capital cost makes the PV investment looks attractive. PV system capital cost was reduced as follows (0.9, 0.8, 0.7, 0.6,

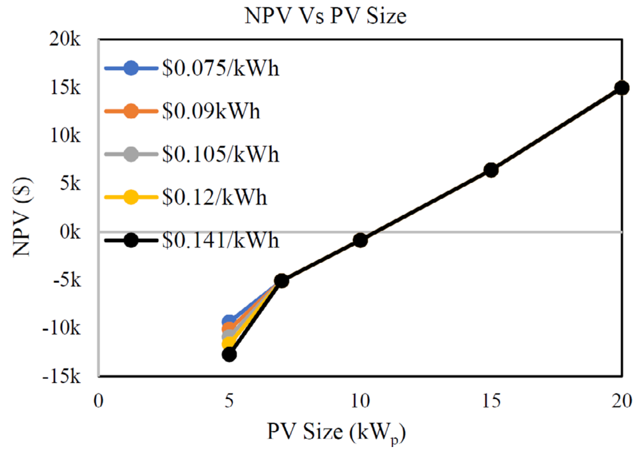

FIGURE 13 Effect of power purchasing prices on the NPV

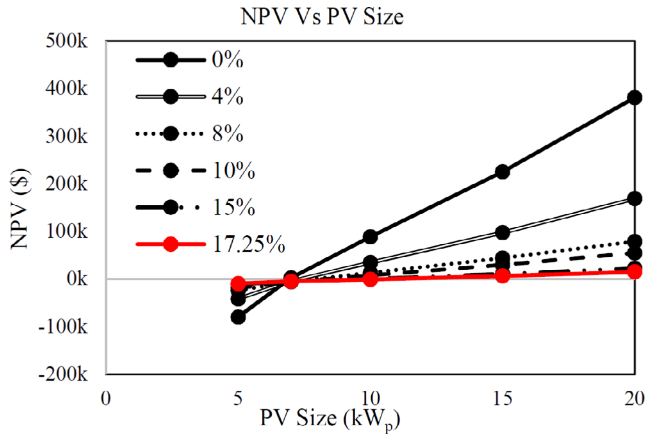

FIGURE 14 Effect of discount rare variations on the NPV

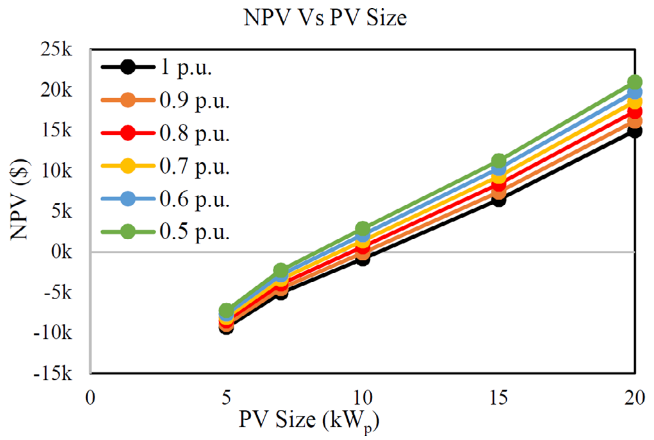

FIGURE 15 Effect of discount rate variations on the NPV

and 0.5) from the original cost. These variation aims to study its effect on the NPV as shown in Figures 11, 15 and 19. These curves prove that, the lower PV system capital cost, the higher the NPV's.

Also, the sensitivity analysis includes the calculation of the boundary values for PBP and IRR at each load pattern.

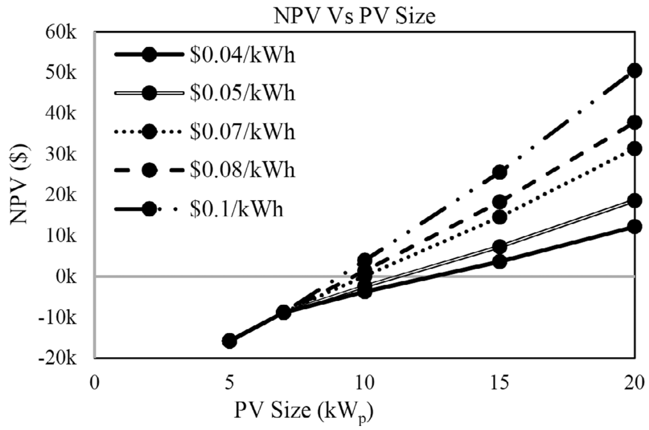

FIGURE 16 Effect of discount rate variations on the NPV

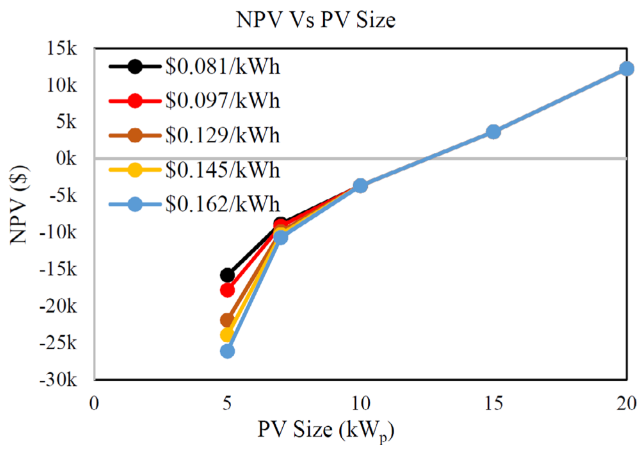

FIGURE 17 Effect of capital cost reduction on the NPV

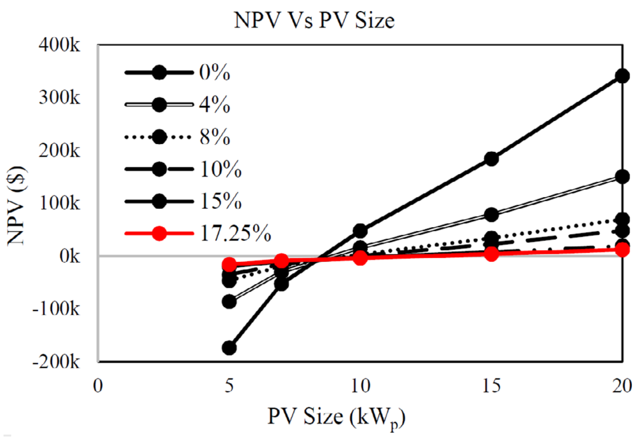

FIGURE 18 Effect of capital cost reduction on the NPV

6.1 | Low energy consumption pattern

The base case for this system is to feed the load (18.8 kWh/d) from the grid with $0.05/kWh power purchasing price and $0.04/kWh sellback price and 17.25% discount rate.

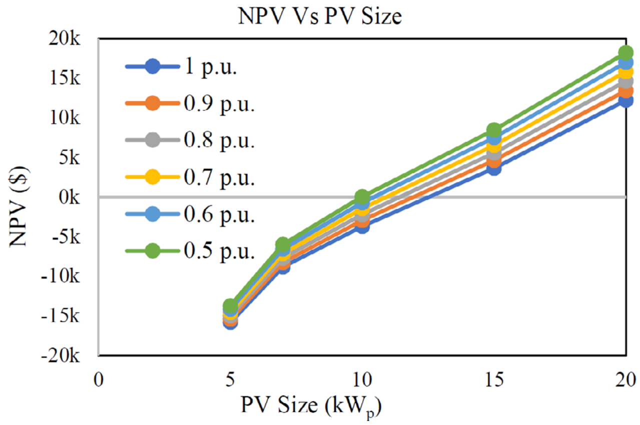

FIGURE 19 Effect of capital cost reduction on the NPV

6.1.1 | Effect of sellback power price variation

For this load pattern, Increasing the sellback price will increase the NPV for each system size, that is, at sellback power price \$0.04/kWh the system will be profitable at 7.8 kW_p, while at \$0.1/kWh the system will be profitable at 5.1 kW_p as shown in Figure 8.

6.1.2 | Effect of subsidy releasing

The tariff was varying as follows; (\$0.05, 0.07, 0.09, 0.1, and 0.123/kWh).

For this load pattern, even 5 kW_p PV system is enough to cover the load requirement, that is, there is no energy purchased from the grid so, there is no effect for increasing the grid purchasing price on the NPV. Unlike Reference 7, which indicated that, increasing the power price will enhance the system profitability while it depends on the energy consumption and the amount of sold and purchased energy to the utility grid as shown in Figure 9.

6.1.3 | Effect of discount rate variation

Increasing the discount rate for each PV size will reduce the NPV and will reduce the system profitability, that is, at 0% discount rate, the system will be profitable at 4.7 kW_p, while at 17.25% the system will be profitable at 7.8 kW_p. Variation of the discount rate, neither affect the payback period nor the internal rate of return as shown in Figure 10.

6.1.4 | Effect of PV system capital cost reductions

For this load pattern and for any system size, decreasing the capital cost of the system will increase the NPV which makes the system more profitable and increase the bill savings that will affect the payback period.

For this load pattern, the minimum value of the PBP is 1.23 years and the maximum value of the IRR is 81% can be achieved at 20 kW_p PV system size, 0.5 p.u. cost (min), 0.123 power purchasing price (max) and 0.1 sellback power price (max), while the maximum value of the PBP is 8.26 years and the minimum value of the IRR is 9.8% can be achieved at 5 kW_p PV system size, 1 p.u. cost (max), 0.05 power purchasing price (min), and 0.04 sellback power price (min) as shown in Figure 11.

6.2 | Medium energy consumption

The base case for this system is to feed the load (28.9 kWh/d) from the grid with \$0.075/kWh power purchasing price and \$0.04/kWh sellback price and 17.25% discount rate.

6.2.1 | Effect of sellback power price variation

For this load pattern, increasing the sellback price will increase the NPV for each system size, that is, at sellback power price \$0.04/kWh the system will be profitable at 10.7 kW_p, while at \$0.1/kWh the system will be profitable at 7.5 kW_p as shown in Figure 12.

6.2.2 | Effect of subsidy releasing

The tariff was varying as follows; (\$0.075, 0.09, 0.105, 0.12, and 0.141/kWh).

For this load pattern, changing the electricity tariff will affect only the 5kW_p PV size NPV as it is not sufficient to feed the load, even 7 kW_p PV system is enough to cover the load requirement, that is, there is no energy purchased from the grid so, there is no effect for increasing the grid purchasing price on the NPV as shown in Figure 13.

6.2.3 | Effect of discount rate variation

Increasing the discount rate for each PV size will reduce the NPV and will reduce the system profitability, that is, at 0% discount rate, the system will be profitable at 6.9 kW_p, while at 17.25% the system will be profitable at 10.7 kW_p. variation of the discount rate, neither affect the payback period nor the internal rate of return as shown in Figure 14.

6.2.4 | Effect of PV system capital cost reductions

For this load pattern, due to the relatively high-power consumption, the effect of capital cost reduction will appear for PV capacity greater than 10kW_p that the system will gain profits, while there is not a big effect on the payback period for different sizes at the same capital cost as shown in Figure 15.

For this load pattern, the minimum value of the PBP is 1.15 years and the maximum value of the IRR is 86.7% can be achieved at 20 kW_p PV system size, 0.5p.u. cost (min), 0.141 power purchasing price (max) and 0.1 sellback power price (max), while the maximum value of the PBP is 5.12 years and the minimum value of the IRR is 18.4% can be achieved at 7 kW_p PV system size, 1p.u. cost (max), 0.075 power purchasing price (min) and 0.04 sellback power price (min).

6.3 | High energy consumption

The base case for this system is to feed the load (35.28 kWh/d) from the grid with \$0.081/kWh power purchasing price and \$0.04/kWh sellback price and 17.25% discount rate.

6.3.1 | Effect of sellback power price variation

For this load pattern, increasing the sellback price will increase the NPV for each system size, that is, at sellback power price \$0.04/kWh the system will be profitable at 12.8 kW_p, while at \$0.1/kWh the system will be profitable at 9.1 kW_p as shown in Figure 16.

6.3.2 | Effect of subsidy releasing

The tariff was varying as follows; (\$0.081, 0.097, 0.129, 0.145, and 0.162/kWh).

For this load pattern, changing the electricity tariff will affect only the 5 and 7 kW_p PV size NPV as it is not sufficient to feed the load, even 10kW_p PV system is enough to cover the load requirement, that is, there is no energy purchased from the grid so, there is no effect for increasing the grid purchasing price on the NPV as shown in Figure 17.

6.3.3 | Effect of discount rate variation

Increasing the discount rate for each PV size will reduce the NPV and will reduce the system profitability, that is, at 0% discount rate, the system will be profitable at 8.6 kW_p, while at 17.25% the system will be profitable at 12.8 kW_p. variation of the discount rate, neither affect the payback period nor the internal rate of return as shown in Figure 18.

6.3.4 | Effect of PV system capital cost reductions

For this load pattern, the reduction on the capital cost will affect the PV capacity with 15 kW_p or greater that the system will be profitable, and the rate of savings will increase. On the other hand, the effect of this reduction in addition to the high electricity purchasing price will encourage customers to invest in the PV as the payback period will decrease rapidly as shown in Figure 19.

For this load pattern, the minimum value of the PBP is 1.06 years and the maximum value of the IRR is 94.4% can be achieved at 20 kW_p PV system size, 0.5 p.u. cost (min), 0.162 power purchasing price (max) and 0.1 sellback power price (max), while the maximum value of the PBP is 4.66 years and the minimum value of the IRR is 20.5% can be achieved at 10kW_p PV system size, 1p.u. cost (max), 0.081 power purchasing price (min) and 0.04 sellback power price (min).

7 | CONCLUSIONS

The Egyptian power sector had been briefly discussed with pointing to the different incentive policies which had been used and the policy which is applied now. This article showed how is the PV output power can be expressed in addition to explaining the meaning of some economic expressions and what is refer to. The proposed analysis aimed to evaluate the visibility of adding rooftop PV system to the grid for the residential customers with different load patterns with different power purchasing price for each pattern. Then some variations had been implemented as a sensitive analysis like variations on the sellback power price, the effect of changing the discount rate and changing the power purchasing tariff.

Based on the data analyzed it was found that, for low energy consumption, the best system size which will gain the minimum COE in addition to the minimum NPC is 20 kW_p PV system that will also give the maximum renewable fraction of 91.1%, while the 5 kW_p PV system will provide 67.5% of renewable fraction. Installing 10kW_p is the profitable PV size as the NPV is \$3450. For medium energy consumption, the best system size which will gain the minimum COE in addition to the minimum NPC is 20 kW_p PV system that will also give the maximum renewable fraction of 86.7%, while the 5 kW_p PV system will provide 55.6% of renewable fraction. Installing 15 kW_p is the profitable PV size as the NPV is \$6448. For high energy consumption, the best system size which will gain the minimum COE in addition to the minimum NPC is 20 kW_p PV system that will also give the maximum renewable fraction of 81.1% and \$10 255 NPV, while the 5kW_p PV system will provide 42.5% of renewable fraction. It is preferred to increase the PV size while increasing the load pattern to reduce the PBP and maximize the bill savings.

For all three different load patterns, variation and changing in the both sellback power price and discount rate will not affect the renewable fraction or the power purchased or sold to the grid, but will effect on the NPC and the COE, while the effect of changing the electricity tariff mainly depends on the load consumption pattern, the PV Size installed and the amount of energy purchased from the grid.

Regarding the input economical Egyptian parameters, results showed that, for all energy consumption patterns and for a certain inflation rate, increasing the discount rate will reduce the profitability of the PV system that may discourage the customers to invest in the PV projects.

According to the yearly decrease in the PV system costs, the effect of this reduction had been studied and it was found that decreasing the system capital cost will encourage investing in PV system especially for the high consumption load pattern that will maximize the savings and reduce the payback period that means the system will gain its profits quickly.

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